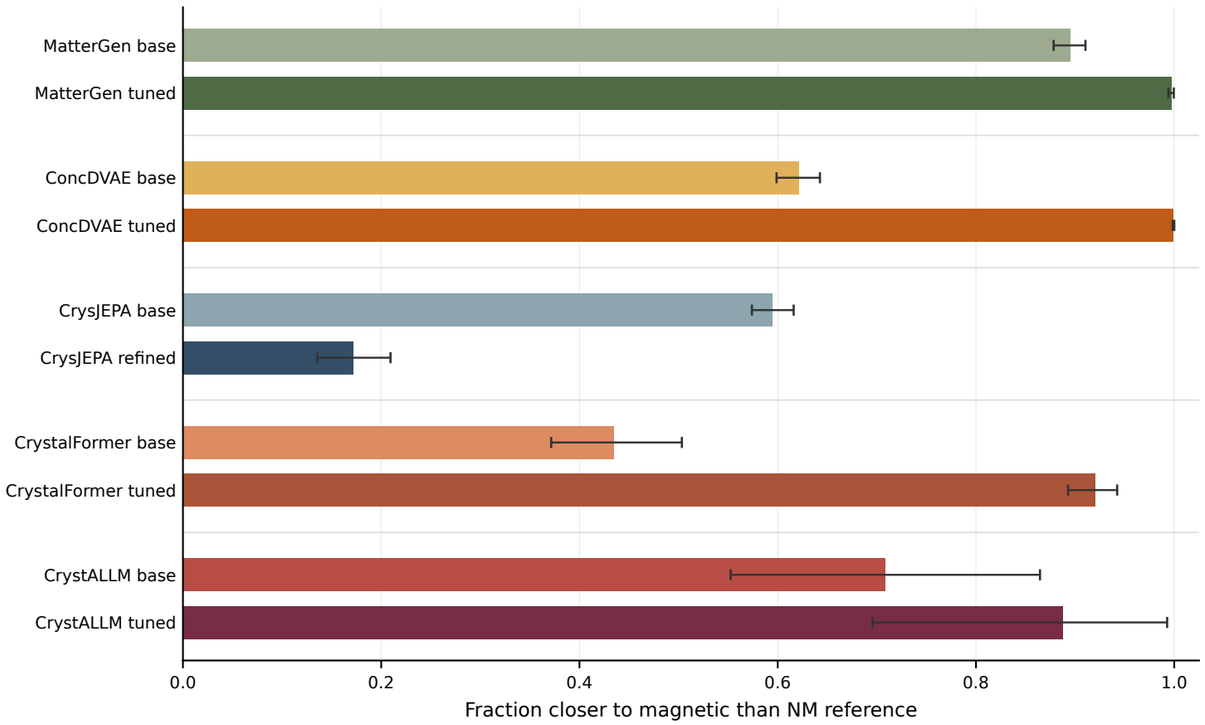
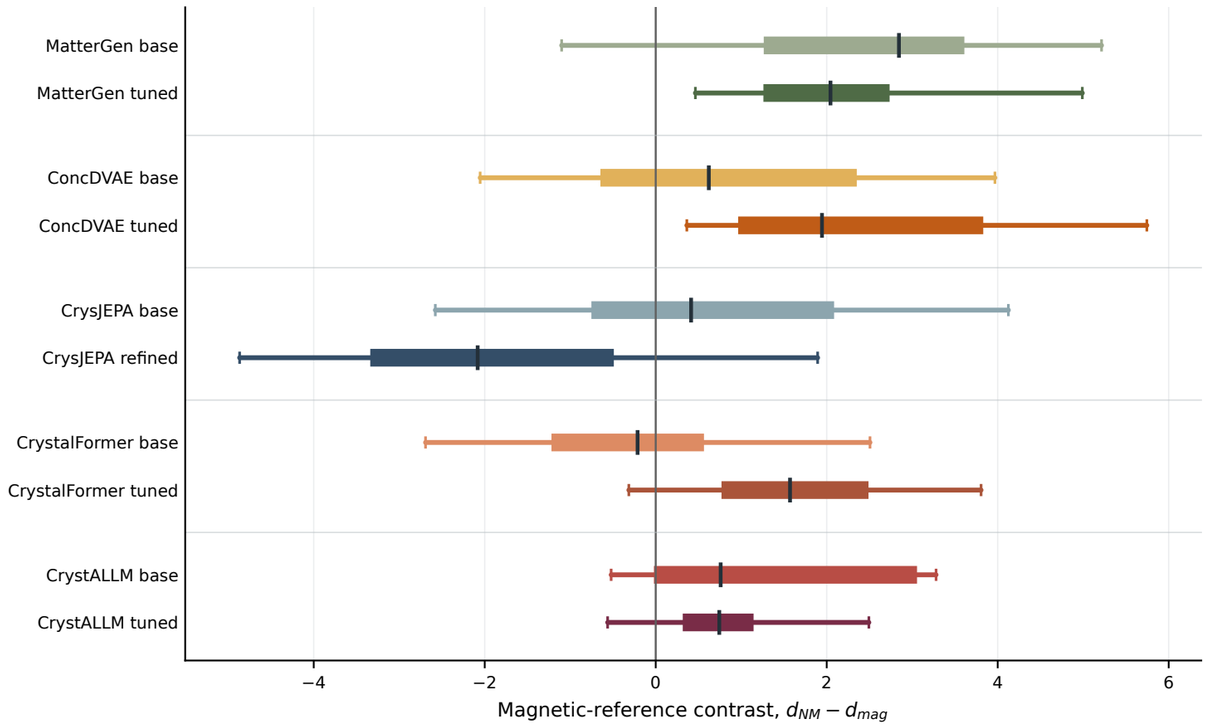
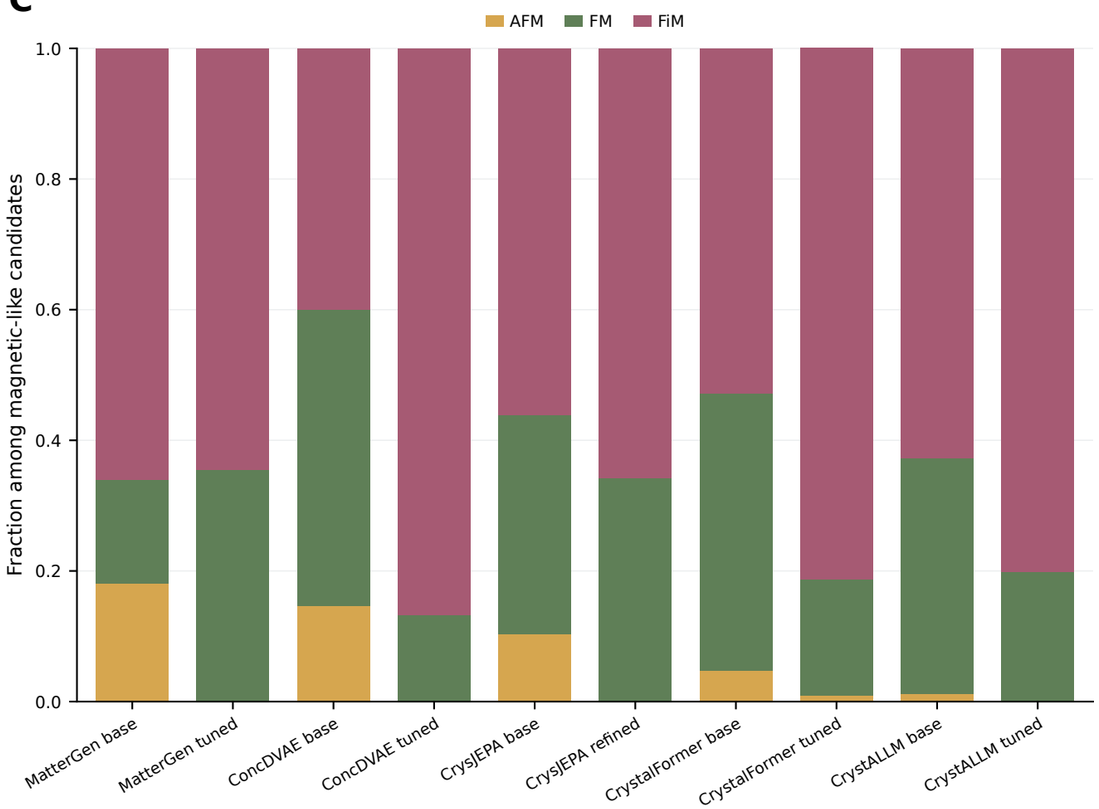
A


MatterGen base
MatterGen tuned
ConcDVAE base
ConcDVAE tuned
CrysJEPA base
CrysJEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned
0.0
0.2
0.4
0.6
0.8
1.0
Fraction closer to magnetic than NM reference

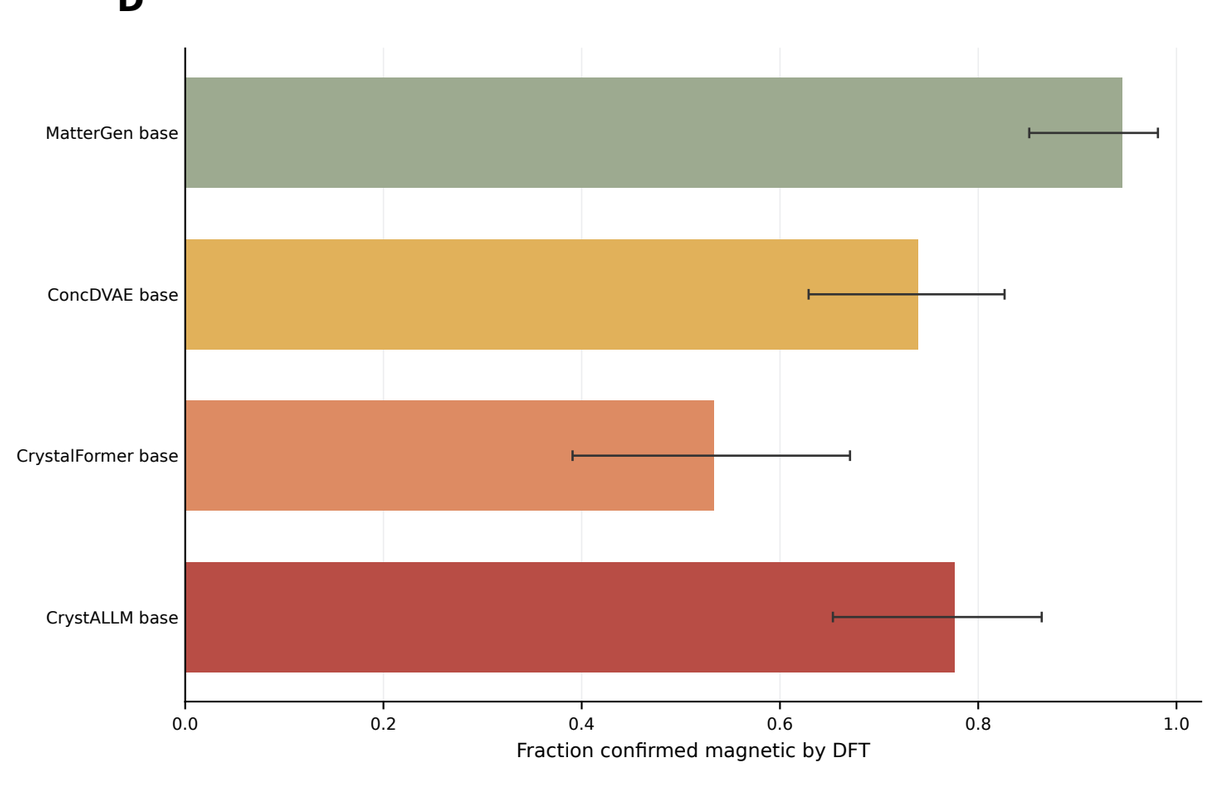
B


MatterGen base
MatterGen tuned
ConcDVAE base
ConcDVAE tuned
CrysJEPA base
CrysJEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned
−4
−2
0
2
4
6
Magnetic-reference contrast, $d_{NM} - d_{mag}$

C


AFM
FM
FiM
Fraction among magnetic-like candidates
0.0
0.2
0.4
0.6
0.8
1.0
MatterGen base
MatterGen tuned
ConcDVAE base
ConcDVAE tuned
CrysJEPA base
CrysJEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned

D


MatterGen base
ConcDVAE base
CrystalFormer base
CrystALLM base
0.0
0.2
0.4
0.6
0.8
1.0
Fraction confirmed magnetic by DFT